ESTERS 9

The various scents of fruits and flowers are caused by the presence of esters; so it should come as no surprise to learn that esters are often used as natural flavourings. However, esters are most commonly used as an important part of industrial solvents.

Esters have two carbon chains separated by an oxygen atom. They have the general formula R–COOR', where R is derived from an organic acid and R' from an alkanol (alcohol). Figure 1 shows the general structure of an ester. Esters can be detected in water in concentrations as low as a few parts per billion (ppb). Esters can be produced by reacting an organic acid with an alkanol (alcohol) in an equilibrium reaction:

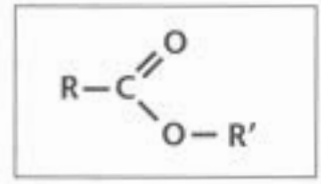


Figure 1

acid + alkanol ⇌ ester + water

Figure 2 shows the general structure of an alkanol. Figure 3 shows the general structure of an alkanoic (carboxylic) acid. The simplest ester is formed from the reaction between methanoic (HCOOH, formic) acid and methanol (CH_3OH). It is called methyl methanoate (methyl formate). Figure 4 shows the structure of methyl methanoate.

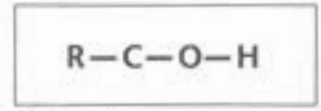


Figure 2

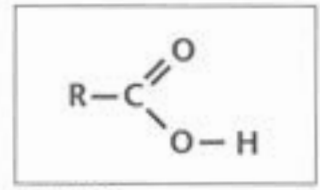


Figure 3

Esterification is a condensation reaction; a molecule of water is formed. The reaction can be speeded up by adding concentrated sulfuric acid which removes the water produced and drives the equilibrium reaction to the right.

1 Figure 5 shows the reaction between methanoic acid and methanol. Copy figure 5 and write a balanced equation for this reaction.

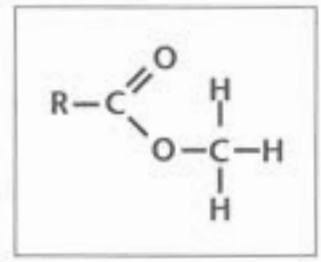


Figure 4

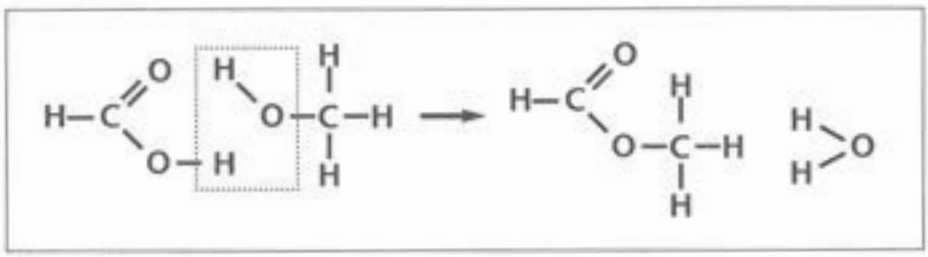


Figure 5

In the names of esters, the alkanol contributes the first part of the name and the acid the second part. For example, alkanoic acids form alkanoates, methanoic acids form methanoates, and ethanoic acids form ethanoates. Note that the name is the 'reverse' of the order in which the acid structure is usually drawn.

2 Using figure 5 as a model, show the reaction between ethanol and ethanoic acid (CH_3CHOOH) forming ethyl ethanoate.

3 Copy and complete the table. The first example has been done for you.

Name of ester	Formula of ester	Parent acid	Parent alkanol	Scent
Methyl butanoate	$C_4H_7OOCH_3$	Butanoic	Methanol	Pineapple
		Ethanoic	Pentanol	Banana
Ethyl methanoate				Raspberry
Octyl ethanoate				Orange

4 What is represented by *ppb*?

5 What is a *condensation* reaction?

6 What is a *fragrance*?

7 Do you think that esters play a role in the making of perfumes? Explain your answer.

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